Euler's Totient over Pairwise Coprime Products

A reading point for Mathlib PR #45343

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Status. Proposed Mathlib contribution, under review. The revised Lean declarations reportedly compile and pass local lint checks; merging into Mathlib remains pending.

Links. https://github.com/leanprover-community/mathlib4/pull/45343

1. What does the totient count?

Euler's totient function $\varphi(n)$ counts the integers in $\{1, \dots, n\}$ whose greatest common divisor with n is 1. For example, the integers coprime to 9 are

$$\{1, 2, 4, 5, 7, 8\}, \qquad \boxed{\varphi(9) = 6}.$$

Two numbers a, b are *coprime* if $\gcd(a, b) = 1$.

2. An existing two-factor theorem

If $\gcd(a, b) = 1$, Euler's totient is multiplicative:

$$\boxed{\varphi(ab) = \varphi(a)\varphi(b)}.$$

Mathlib already provides this fact as `Nat.totient_mul`. With $a = 4$ and $b = 9$:

$$\gcd(4, 9) = 1, \quad \varphi(4) = 2, \quad \varphi(9) = 6, \quad \varphi(36) = 12 = 2 \cdot 6.$$

3. The proposed general theorem

Suppose s is a finite collection of indices and $f(i) \in \mathbb{N}$ is a factor for each index $i \in s$. The *pairwise-coprime hypothesis* says

$$\forall i, j \in s, \qquad i \neq j \Longrightarrow \gcd(f(i), f(j)) = 1.$$

Then the new theorem asserts

$$\boxed{\varphi\left(\prod_{i \in s} f(i)\right) = \prod_{i \in s} \varphi(f(i))}.$$

Its proposed Lean name is `Nat.totient_prod_of_pairwise_coprime`. The indexed form includes an ordinary finite set of numbers by taking $f(i) = i$.

For a concrete reading point, 4, 9, and 25 are pairwise coprime, even though each is composite:

$$\varphi(4 \cdot 9 \cdot 25) = \varphi(900) = \varphi(4)\varphi(9)\varphi(25) = 2 \cdot 6 \cdot 20 = \boxed{240}.$$

4. Why induction proves it

1. **Empty collection:** the empty product is 1, and $\varphi(1) = 1$.
2. **Induction hypothesis:** assume the product identity holds for the factors already in the collection.
3. **Add a factor $f(a)$, with $a \notin s$:** pairwise coprimality makes $f(a)$ coprime to each old factor, hence to their entire product.
4. **Apply multiplicativity:** use `Nat.totient_mul` to separate $\varphi(f(a) \cdot \prod_{i \in s} f(i))$.
5. **Finish:** substitute the induction hypothesis for the old product.

The Lean proof uses finset induction and existing coprimality lemmas to justify every step.

5. Distinct primes as a consequence

For a prime p, $\varphi(p) = p - 1$. Also, two *distinct* primes are coprime. A finite set contains each element only once, so if every member of a finite set s is prime, the general theorem yields

$$\boxed{\varphi\left(\prod_{p \in s} p\right) = \prod_{p \in s} (p - 1)}.$$

The requested Mathlib name is `Nat.totient_prod_of_forall_prime`. The prime corollary is declared as its own theorem for downstream use, but its proof invokes the general result rather than repeating the induction.

For $s = \{2, 3, 5\}$:

$$\varphi(30) = (2 - 1)(3 - 1)(5 - 1) = \boxed{8}.$$

The eight residue representatives coprime to 30 are

$$\boxed{\{1, 7, 11, 13, 17, 19, 23, 29\}}.$$

This count connects the totient result to the mod-30 reading lanes used in the General Divisor Theorem work. The residue condition $\gcd(n, 30) = 1$ alone does *not* assert that n is prime.

6. What the Mathlib contribution adds

The mathematical identity is classical. The proposed contribution makes the *finite-product form* directly reusable in Mathlib, then derives the prime-product identity from it. Previously, users could reconstruct these consequences from two-factor multiplicativity and finset induction.

The revision was requested during review of PR #45343. Claude assisted with Lean code and local checks; ChatGPT and DeepSeek assisted with review. The contributor directs the work and is responsible for understanding and maintaining the submitted proof. The reported local build and lint checks succeeded; this verification is distinct from Mathlib's decision to merge the change.

Research context: translatepi.ai/main.pdf
Mathlib PR: leanprover-community/mathlib4#45343